

A novel mechanism for bacterial sporulation based on programmed peptidoglycan degradation

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eLife Assessment

This **important** study identifies and partially characterises two proteins optimised for coordinated peptidoglycan degradation during two spore morphogenesis programs in the bacterium *Myxococcus xanthus*. The evidence supporting the conclusions is **solid**, although the description of the data is somewhat overstated. After some editing, the paper will be of interest to those studying peptidoglycan synthesis and reorganisation, which is a central aspect of microbial cell biology.

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Abstract

Many bacteria form spores to endure unfavorable conditions. While *Firmicutes* generate endospores through cell division, sporulation in non-Firmicutes remains less understood. The Gram-negative bacterium *Myxococcus xanthus* undergoes sporulation through two distinct mechanisms: rapid sporulation triggered by chemical induction and slow sporulation driven by starvation, both occurring independently of cell division. Instead, these processes depend on the complete degradation of the peptidoglycan (PG) cell wall by lytic transglycosylases (LTGs), with both LtgA and LtgB supporting rapid sporulation and LtgB alone driving slow sporulation. Remarkably, LtgB programs the pace of PG degradation by LtgA during rapid sporulation, ensuring a controlled process that prevents abrupt PG breakdown and the formation of non-resistant pseudospores. In addition to regulation between LTGs, PG degradation is also influenced by its synthesis; cells exhibiting increased mureptide production often circumvent sporulation. These findings not only reveal novel mechanisms of bacterial sporulation but also shed light on the regulatory network governing PG dynamics.

Introduction

Spores are metabolically dormant cells that can survive unfavorable conditions, including extremes of temperature, desiccation, and ionizing radiation (Huang & Hull, 2017 [↗](#), Hutchison *et al.*, 2014 [↗](#)). Sporulation, the process of spore development, is a strategy utilized by a wide variety of organisms, from bacteria and protozoa to fungi and plants. In addition to protective spore coats, spore resilience relies on cytosol dehydration and DNA compaction—key processes that decrease cell volume and alter cell shape (Higgins & Dworkin, 2012 [↗](#)). Thus, morphological differentiation is a hallmark of sporulation. In bacterial spore formers, as peptidoglycan (PG) cell walls largely determine cell morphology, their sporulation requires profound PG remodeling.

PG is a rigid, mesh-like macromolecule that is composed of glycan strands of repeating units of N-acetyl glucosamine (GluNAc)-N-acetyl muramic acid (MurNAc) crosslinked by peptides (Egan *et al.*, 2020 [↗](#)). PG encloses the entire cytoplasmic membrane, and its rigidity provides mechanical support against osmotic stress. For this reason, PG is an essential structure for most bacteria and a major target for antibacterial treatments. Sporulation provides invaluable opportunities for understanding the dynamics of PG, which is controlled by multiple synthases and hydrolases. PG synthases include glycosyltransferases (GTases) that polymerize glycan strands and transpeptidases (TPases) that form peptide crosslinks (Egan *et al.*, 2020 [↗](#)). PG hydrolases, also known as autolysins, include glycosidases that break the glycan strands, and amidases and endo/carboxypeptidases that cleave the peptide crosslinks (Egan *et al.*, 2020 [↗](#), Rohs & Bernhardt, 2021 [↗](#), van Heijenoort, 2011 [↗](#)).

Firmicutes, including Gram-positive bacteria like *Bacilli* and *Clostridia*, produce endospores. Their morphological transition from rod-shaped vegetative cells to ovoid spores occurs through asymmetric cell division, resulting in the formation of a smaller forespore and a larger mother cell. Eventually, the forespore becomes an oval endospore after being engulfed by the mother cell (Higgins & Dworkin, 2012 [↗](#), McKenney & Eichenberger, 2012 [↗](#)). A mature endospore contains two PG layers: the germ cell wall, derived from the forespore's original PG, and a thickened PG cortex deposited by the mother cell (Popham & Bernhardt, 2015 [↗](#)). While a few Gram-negative spore formers also belong to the *Firmicutes* phylum, they share conserved sporulation genes with *Bacilli* and *Clostridia*, which suggests similar sporulation mechanisms (Tocheva *et al.*, 2011 [↗](#), Yutin & Galperin, 2013 [↗](#)).

Sporulation by non-firmicutes bacteria has been largely overlooked. Myxobacteria, a group of Gram-negative, non-firmicutes spore formers, were first assigned to the phylum δ -proteobacteria, but recently reclassified into the newly established phylum *Myxococcota* (Waite *et al.*, 2020 [↗](#)). *Myxococcus xanthus* is a model organism of myxobacteria. Rod-shaped *M. xanthus* cells can undergo sporulation via two distinct pathways, both resulting in spherical spores. First, in response to certain chemical signals, such as glycerol and dimethyl sulfoxide (DMSO), individual *M. xanthus* cells can rapidly transition into isolated spherical spores in aqueous environments within hours (Dworkin & Gibson, 1964 [↗](#)). Second, millions of cells can aggregate on solid starvation media and develop into spore-filled fruiting bodies, a process that takes a few days to complete (O'Connor & Zusman, 1988 [↗](#)). Both sporulation pathways are programmed, tightly controlled processes that involve over 1,000 genes (Muller *et al.*, 2010 [↗](#), Munoz-Dorado *et al.*, 2019 [↗](#)). In contrast to endospore formation, cell division is not involved in either the *M. xanthus* sporulation mechanisms, and *M. xanthus* lacks the sporulation genes in *Firmicutes* (Aramayo & Nan, 2022 [↗](#)). Thus, sporulating *M. xanthus* must accomplish the rod-to-sphere transition through yet to be discovered PG remodeling mechanisms, which provide invaluable opportunities to investigate PG dynamics (Zhang *et al.*, 2021 [↗](#)).

Distinct from *Bacilli* and *Clostridia*, *M. xanthus* spores lack cortex PG (Bui *et al.*, 2009 [\[1\]](#), Zhang *et al.*, 2020 [\[2\]](#), Voelz & Dworkin, 1962 [\[3\]](#)). Rather, their resistance is derived from polysaccharide coats (Kottel *et al.*, 1975 [\[4\]](#), Perez-Burgos *et al.*, 2020 [\[5\]](#), Saidi *et al.*, 2021 [\[6\]](#), Muller *et al.*, 2012 [\[7\]](#)). Sporulating cells that fail to deposit coat polysaccharides on their surfaces produce spherical pseudospores that lack resistance (Wartel *et al.*, 2013 [\[8\]](#), Zhang *et al.*, 2020 [\[2\]](#), Holkenbrink *et al.*, 2014 [\[9\]](#)). In a pioneering study, Bui *et al.* did not detect muropeptides in glycerol-induced spores, indicating that *M. xanthus* degrades its PG during rapid sporulation (Bui *et al.*, 2009 [\[1\]](#)). However, it remains unclear whether starvation-induced spores within fruiting bodies still retain PG and how *M. xanthus* orchestrates programmed morphological changes.

In this report, we used ultra-performance liquid chromatography (UPLC) to demonstrate that *M. xanthus* degrades PG in both sporulation pathways. Through mutagenesis studies, we discovered that these sporulation pathways require different lytic transglycosylases (LTGs) for PG degradation. While the rapid sporulation relies on both LtgA and LtgB, the slow formation of fruiting body spores only requires LtgB. Remarkably, LtgB regulates the pace of PG degradation by LtgA during rapid sporulation, preventing the formation of non-resistant pseudospores due to abrupt PG breakdown. This research not only uncovers novel mechanisms of sporulation in non-firmicutes but also highlights the crucial role of cross-regulation between PG hydrolases in maintaining cell integrity.

Results

PG is degraded in both spore types

To investigate if starvation-induced spores in *M. xanthus* fruiting bodies retain PG, we broke the fruiting bodies after 120 h of starvation and purified sacculi from spores. For comparison, we also purified the sacculi from vegetative cells and glycerol-induced spores. PG contents in all three cell types were analyzed using UPLC after muramidase digestion (see materials and Methods). The two spore types showed similar profiles of a discernible presence of muropeptides that resembled those found in vegetative cells, albeit in significantly reduced quantities (Fig. 1 [\[1\]](#)). Importantly, the most abundant muropeptides identified in the vegetative cells, MurNAc-tetrapeptide monomer (M4) and MurNAc-tetrapeptide dimer (D44), were nearly absent in both spore types (Fig. 1 [\[1\]](#)). Such residual muropeptides in both spore types are inadequate for forming continuous PG layers (Bui *et al.*, 2009 [\[1\]](#)). This observation indicates that spore development in both pathways involves the breakdown of the vegetative PG cell wall.

Despite the overall decline in muropeptides, anhydro-muropeptides markedly increased in both spore types, comprising over 90% of the total muropeptides (Fig. 1 [\[1\]](#)). Especially, anhydro-MurNAc-tetrapeptide trimer (T444^{Anh}) that was only detected in trace amounts in vegetative cells, became a prominent muropeptide in both spore types (Fig. 1 [\[1\]](#)). Moreover, new anhydro-muropeptides, including anhydro, anhydro-MurNAc-tetrapeptide dimer (D44^{Anh,Anh}), anhydro, anhydro-MurNAc-tetrapeptide trimer (T444^{Anh,Anh}), and an anhydrodimer without a NAG (D44^{Anh}(-NAG)), were only present in spores (Fig. 1 [\[1\]](#)). Because anhydro-muropeptides are the signature products of LTGs (Dik *et al.*, 2017 [\[2\]](#), Williams *et al.*, 2018 [\[3\]](#)), their abundance in spores indicates that certain LTGs must play essential roles in *M. xanthus* sporulation.

The two sporulation pathways require different LTGs

The genome of *M. xanthus* encodes 14 putative LTGs (Aramayo & Nan, 2022 [\[4\]](#), Ramirez Carbo *et al.*, 2024 [\[5\]](#)). We imaged the cells of the 14 knockout mutants (Ramirez Carbo *et al.*, 2024 [\[5\]](#)) at different time points after glycerol induction and used their length/width (L/W) ratios to monitor potential defects in the sporulation process. Wild-type cells initiated sporulation within 1 h of glycerol induction, which was evidenced by the increase of cell width and decrease of cell length (Fig. 2A [\[6\]](#)). Consistent with the change in cell morphology, their L/W ratios decreased continuously and

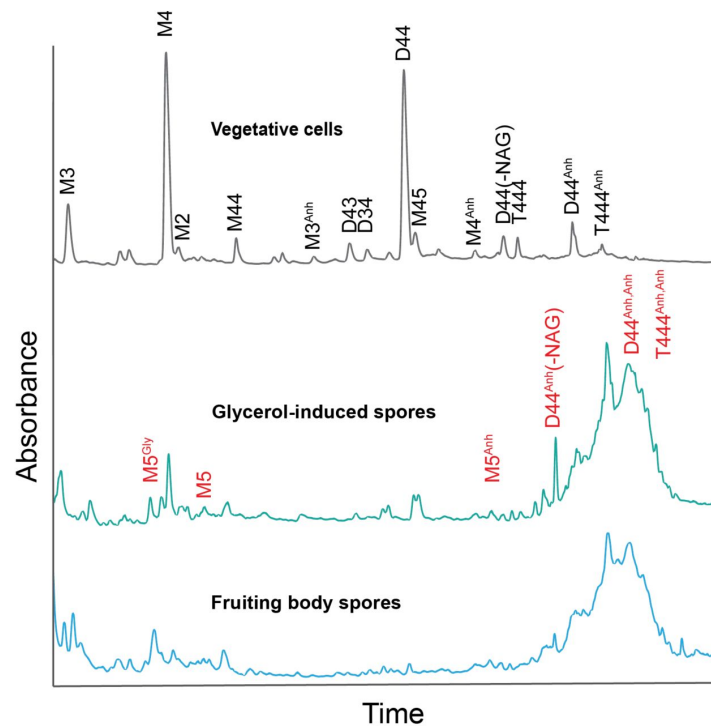


Fig. 1.

Mature *M. xanthus* spores induced by either glycerol or starvation do not contain significant mucopeptides.

UPLC mucopeptides profiles of vegetative cells, glycerol- and starvation-induced spores indicate that the major mucopeptides species, M4 and D44, in vegetative cells are diminished in both spore types, while spores are enriched in anhydro-mucopeptides (Anh). The characteristic peaks are labeled as follows: M, monomeric mucopeptide (uncrosslinked); D, dimeric mucopeptide (crosslink connecting two mucopeptides), T, trimeric mucopeptide (crosslink connecting three mucopeptides). Numbers refer to the status of the peptide side chain (3, tripeptide; 4, tetrapeptide). Red characters mark the mucopeptides only detected in spores.

stabilized after 2 h of induction, when sporulation completed (**Fig. 2B**). In contrast, the cells that carry the deletion of open reading frame (ORF) K1515_20820 (MXAN_RS16290) (Aramayo & Nan, 2022) were able to shorten cell length slightly but retained rod shape after prolonged induction.

Surprisingly, another mutant that carries the deletion of ORF K1515_17460 (MXAN_RS19615) started to lose rod shape immediately after glycerol induction and became spherical within 1 h (**Fig. 2A, 2B**). However, different from the wild-type spores that appeared dark and heterogeneous under differential interference contrast (DIC) microscopy, the spheres formed by this mutant appeared bright and homogenous, similar to the pseudospores from the *ΔaglQS* mutant that lack the motor for depositing spore coat polysaccharides onto cell surfaces (Zhang *et al.*, 2020) (**Fig. 2A**). To test whether these spheres are real spores, we subjected them to sonication and quantified their survival rate using a Helber bacterial counting chamber. After sonication, while $91 \pm 6\%$ (calculated from three independent experiments, $n > 1,000$, same below) of the wild-type spores appeared intact, only $3 \pm 1\%$ of the spheres produced by the K1515_17460 deletion mutant remained. Thus, this mutant indeed formed pseudospores that lacked the resistance against sonication. As the products of both K1515_20820 and K1515_17460 show homology to MltE, an LTG in *Escherichia coli* (encoded by *emtA*, **Fig. 2D**), we named them *LtgA* and *LtgB*, respectively. Both *LtgA* and *LtgB* are required for forming resistant spores via the rapid sporulation pathway, albeit playing opposite roles in the rod-to-sphere transition.

To identify the LTGs that are required for slow sporulation, we grew the 14 knockout strains in rich liquid media and spotted cells on solid starvation (CF) media. After 96 h of incubation, when wild-type cells formed dark fruiting bodies on the agar surface, the *ΔltgB* strain only formed flat aggregations (**Fig. 2C**). To test if *ΔltgB* cells could generate starvation-induced spores without forming fruiting bodies, we scraped such cell aggregations from the agar surface, suspended cells in rich liquid media, subjected them to sonication, then plated them on solid rich media. These cells failed to form colonies after five days of incubation, indicating that *LtgB* is essential for forming starvation-induced spores. In contrast, deleting *LtgA* did not affect fruiting body formation (**Fig. 2C**). Thus, the slow sporulation pathway only requires *LtgB*.

LtgA and LtgB play distinct roles in different sporulation pathways

Do *LtgA* and *LtgB* degrade PG during sporulation? To answer this question, we purified cell sacculi and imaged PG using immunofluorescence and an anti-PG serum (de Pedro *et al.*, 1997). The sacculi of vegetative cells from the wild-type, *ΔltgA*, and *ΔltgB* strains were not visible under DIC microscope, likely due to their flattened structures minimizing light interference. However, PG from all three strains was readily detected in the fluorescence channel (**Fig. 3A**). After 6 h of glycerol-induction, the sacculi of both the wild-type and *ΔltgA* cells remained visible under DIC microscope (**Fig. 3A**), likely due to the deposition of spore coat polysaccharides that sustained unflattened cell structures. After the sacculus purification process, only background PG signals were detected in glycerol-induced wild-type spores (**Fig. 3A**). In contrast, the *ΔltgA* cells retained PG in substantial quantities (**Fig. 3A**). Sacculi from the *ΔltgB* pseudospores only contained a small amount of PG (**Fig. 3A**). Thus, compared to *LtgA*, the enzymatic activity of *LtgB* plays a more limited role in PG degradation during rapid sporulation. Different from the induced wild-type and *ΔltgA* cells, sacculi from the *ΔltgB* pseudospores were undetectable under DIC microscopy (**Fig. 3A**), reflecting the absence of rigid polysaccharide coats. Consequently, the residual PG in these pseudospores lost integrity during purification, with many sacculi displaying irregular shapes and ruptures in the fluorescence channel (**Fig. 3A**). Collectively, *LtgB* appears to be a pace-keeper that prevents abrupt PG degradation and thus allows sporulating cells to assemble polysaccharide coats during rapid sporulation.

To determine the roles of *LtgA* and *LtgB* in slow sporulation, we first scraped the wild-type and *ΔltgA* fruiting bodies from CF agar surface (**Fig. 2C**), dispersed spores by sonication, purified their sacculi, and visualized PG using immunofluorescence. Spherical spores from both the wild-type and *ΔltgA* cells were visible under DIC microscope, indicating that they retained unflattened

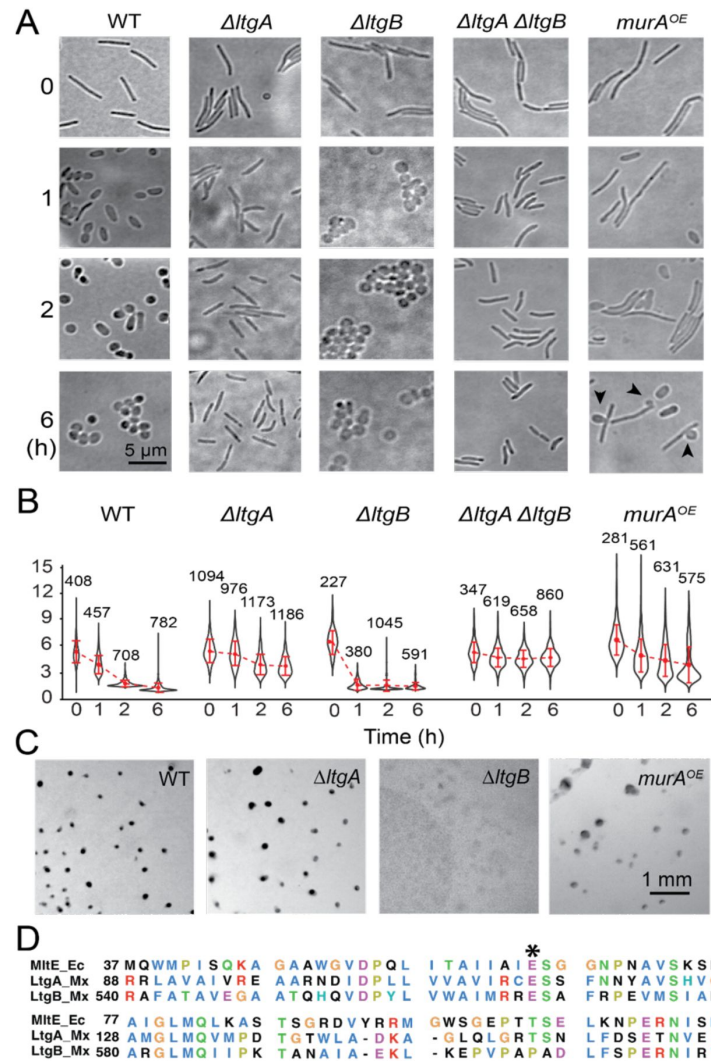


Fig. 2.

Glycerol-induced sporulation is regulated by both LtgA and LtgB, while starvation-induced sporulation only requires LtgB.

A) Bright field images of cells at different time points after glycerol-induction. Black arrows point to lysing cells. **B)** Quantitative analysis of glycerol-induced sporulation using the length/width ratio (L/W) of cells. Whiskers indicate the 25th – 75th percentiles and red dots the median. The total number of cells analyzed is shown on top of each plot. **C)** Only LtgB is required for fruiting body formation on starvation agar. **D)** LtgA and LtgB are homologous to *E. coli* MltE. The asterisk marks the conserved active site.

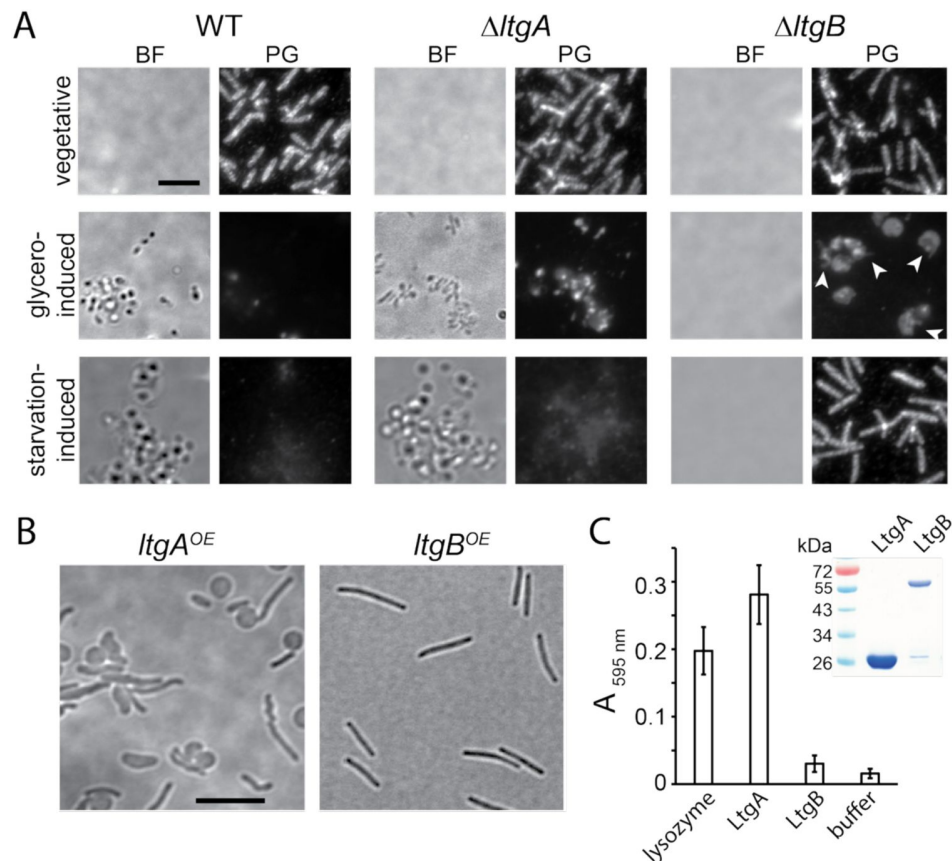


Fig. 3.

LtgA and LtgB play distinct roles in the two sporulation pathways.

A) While LtgA is required for PG degradation during glycerol-induced sporulation, LtgB is the major LTG for forming starvation-induced spores. PG was detected using an anti-PG serum and a fluorescence-conjugated secondary antibody. PG sacculi were purified from cells after 6 h and 120 h of glycerol-induced and starvation-induced sporulation, respectively. Flattened sacculi are not visible under bright field microscopy. White arrows point to the spherical cells that show disintegrated PG. BF, bright field. **B)** The overexpression of LtgA, but not LtgB, collapses rod-shape in vegetative cells. **C)** Purified LtgA and LtgB solubilize dye-labeled PG sacculi at different rates. Lysozyme and buffer serve as the positive and negative controls, respectively. Absorption at 595 nm was measured after 18 h incubation at 25 °C. Data are presented as mean values $\pm$ SD from three technical replicates. The inset shows purified LtgA and LtgB in a Coomassie stained gel. Scale bars, 5 μ m.

shapes. These fruiting body spores lacked PG-specific fluorescence (**Fig. 3A** [↗](#)), likely due to both the absence of PG (**Fig. 1** [↗](#)) and the thickened polysaccharide coats (Voelz & Dworkin, 1962 [↗](#)) preventing PG antibody access. Second, we investigated if *ΔltgB* cells degraded PG during starvation-induced sporulation. Because the flat aggregates formed by *ΔltgB* cells on CF agar did not contain mature spores, we scrapped cell aggregates from agar surface and purified their sacculi without sonication. Instead of forming spores, most of the *ΔltgB* cells were still rod-shaped, indistinguishable from vegetative ones (**Fig. 3A** [↗](#)). These observations indicate that LtgB is the only essential LTG for PG degradation during slow sporulation.

The two sporulation pathways vary greatly in duration: glycerol-induced rapid sporulation is completed within two hours, while starvation-induced slow sporulation unfolds over several days. Based on this distinction, we hypothesized that LtgA and LtgB could degrade PG at different rates. We used a vanillate-inducible promoter (Iniesta *et al.*, 2012 [↗](#)) to overexpress LtgA as a merodiploid. Induced by 200 μM vanillate, cells showed heterogeneous morphology, reflecting the variation in LtgA production. Many cells lost rod shape even in the absence of glycerol, indicating that these cells over-degraded their PG by LtgA (**Fig. 2A** [↗](#), **3B** [↗](#)). In contrast, the overexpression of LtgB using the same method did not affect the morphology of vegetative cells (**Fig. 3B** [↗](#)).

To further investigate the activities of these LTGs, we expressed the periplasmic domains of wild-type LtgA (amino acids 21 – 243) and LtgB (amino acids 26 – 710) in *E. coli* (**Fig. 3C** [↗](#)). We purified PG from wild-type *M. xanthus* cells, labeled it with Remazol brilliant blue (RBB), and tested if the purified AgmT variants hydrolyze labeled PG *in vitro* and release the dye (Jorgenson *et al.*, 2014 [↗](#), Ramirez Carbo *et al.*, 2024 [↗](#), Uehara *et al.*, 2010 [↗](#)). Wild-type LtgA solubilized dye-labeled PG, which absorbed light at 595 nm, demonstrating stronger hydrolytic activity than lysozyme, an enzyme that specifically cleaves β-1,4-glycosidic bonds in PG. In contrast, PG incubated with LtgB only showed minimum release of the dye, indicating slow PG hydrolysis (**Fig. 3C** [↗](#)). While we cannot rule out the possibility that our purification and reaction conditions were suboptimal for LtgB, both the *in vitro* RBB assay and the *in vivo* phenotype resulting from LtgB overexpression support our hypothesis that LtgA degrades PG at significantly higher rates than LtgB.

LtgB regulates LtgA during glycerol-induced sporulation

In line with its function in PG degradation, *ltgA* transcription increases approximately twofold during rapid sporulation but remains unchanged in slow sporulation. In contrast, *ltgB* shows a slight increase in expression during slow sporulation but remains unchanged in rapid sporulation (Muller *et al.*, 2010 [↗](#), Munoz-Dorado *et al.*, 2019 [↗](#)). However, transcriptomic data alone do not account for the opposing roles of LtgA and LtgB in the rod-to-sphere transition during rapid sporulation. The resemblance in morphology between glycerol-induced *ΔltgB* cells and uninduced cells overproducing LtgA suggests that the slower-acting LtgB may offset the rapid activity of LtgA during sporulation, allowing sufficient time for spore coat polysaccharide deposition.

To test if LtgB regulates LtgA, we constructed a *ΔltgA ΔltgB* double deletion strain. Cells from this strain shortened their length slightly but failed to abolish rod shape after prolonged glycerol induction, phenocopying the sporulation defect of the *ΔltgA* strain (**Fig. 2A**, **2B** [↗](#)). Hence, LtgB is a regulator upstream of LtgA.

LtgB exhibits a response to glycerol induction earlier than LtgA

Biochemical reactions on PG are unique for the stark size difference between the enzymes and their substrates. While the enzymes are in nanometer scales, their substrates, the PG sacculi, span several micrometers. Under microscope, PG remains stationary but PG-related enzymes are free to move. Even for *M. xanthus* that moves on surfaces, PG-related enzymes move at least two orders of magnitude faster than cell/PG (Zhang *et al.*, 2023b [↗](#), Ramirez Carbo *et al.*, 2024 [↗](#)). Thus, when diffusive enzymes bind to PG, their mobility decreases (Lee *et al.*, 2016 [↗](#), Zhang *et al.*, 2023b [↗](#)). By tracking single fluorescently-labeled enzyme particles, we can approximate their PG-binding in

different physiological conditions and genetic backgrounds (Ramirez Carbo *et al.*, 2024 [↗](#), Zhang *et al.*, 2023b [↗](#)). We individually expressed photo-activatable mCherry (PAmCherry)-labeled LtgA and LtgB using their native loci and promoters. Both labeled LTGs accumulated as full-length proteins (Fig. S1A [↗](#)) and the PAmCherry tags did not affect the formation of either glycerol- or starvation-induced spores, indicating that these fusion proteins were fully functional (Fig. S1B, C [↗](#)). Consistent to the transcriptomic data (Muller *et al.*, 2010 [↗](#), Munoz-Dorado *et al.*, 2019 [↗](#)), vegetative cells produce significantly less LtgA than LtgB (Fig. S1A [↗](#)).

We used a 405-nm excitation laser (0.3 kW/cm^2 , 0.1 s) to activate the fluorescence of a few labeled LTG particles randomly in each cell and quantified their localization using a 561-nm laser at 10 Hz using single particle tracking photo-activated localization microscopy (sptPALM, see Materials and Methods) (Fu *et al.*, 2018 [↗](#), Nan *et al.*, 2015 [↗](#), Nan *et al.*, 2013 [↗](#)). As free PAmCherry particles diffuse extremely fast in the cytoplasm, entering and exiting the focal plane frequently, they usually appear as blurry objects that cannot be followed at 10 Hz close to the cell surface (Fu *et al.*, 2018 [↗](#), Zhang *et al.*, 2023b [↗](#)). For this reason, the noise from free PAmCherry due to potential protein degradation was negligible.

Single-particles of PAmCherry-labeled LtgA and LtgB can be categorized into two populations, immobile and mobile. The immobile particles remained within a single pixel ($160 \text{ nm} \times 160 \text{ nm}$) before photobleach, and the mobile ones displayed typical diffusion (Fig. 4A [↗](#)). For the particles that switched between mobile and immobile states, our algorithm categorized them as mobile and calculated their diffusion coefficients (D) from their entire trajectories that contained both mobile and immobile segments. Hence, binding to PG not only increases the immobile population of the enzyme particles but also decreases the D of the mobile particles. In vegetative cells where large-scale PG degradation does not occur, 23.9% ($n = 1175$) and 18.2% ($n = 824$) of LtgA and LtgB particles were immobile, respectively (Fig. 4B [↗](#)). D values of the mobile population are $2.62 \times 10^{-2} \pm 2.0 \times 10^{-3} \mu\text{m}^2/\text{s}$ ($n = 894$) for LtgA-PAmCherry and $2.80 \times 10^{-2} \pm 2.50 \times 10^{-3} \mu\text{m}^2/\text{s}$ ($n = 674$) for LtgB-PAmCherry (Fig. 4C [↗](#)).

We then determined how the dynamics of both LTGs respond to glycerol induction in the rapid sporulation pathway. Immediately (1 min) after adding glycerol, the dynamics of LtgA remained little changed, when 21.0% ($n = 928$) of particles were immobile and the D of the mobile ones was $2.53 \times 10^{-2} \pm 0.19 \times 10^{-3} \mu\text{m}^2/\text{s}$ ($n = 733$) (Fig. 4B, 4C [↗](#)). In contrast, the mobility of LtgB decreased significantly, with the immobile population increased to 28.0% ($n = 1359$) and D of the mobile population decreased to $2.17 \times 10^{-2} \pm 2.75 \times 10^{-3} \mu\text{m}^2/\text{s}$ ($n = 978$) (Fig. 4B, 4C [↗](#)). Therefore, these results indicate that upon glycerol induction, LtgB rapidly enhances its binding to PG.

We then chose 30 min after glycerol induction as a time point for rapid PG degradation, which is reflected by the dramatic decrease of L/W during the first hour of sporulation (Fig. 2A [↗](#), 2B). Compared to its inert response 1 min after glycerol induction, the mobility of LtgA particles decreased significantly at 30 min, when the immobile population increased to 31.5% ($n = 1163$) and D of the mobile population decreased to $1.80 \times 10^{-2} \pm 2.12 \times 10^{-3} \mu\text{m}^2/\text{s}$ ($n = 797$) (Fig. 4B, 4C [↗](#)). The reduced mobility of LtgA aligns with its function as the primary LTG in glycerol-induced sporulation. Strikingly different from LtgA, as sporulation advanced to the 30-min mark, LtgB's mobility returned to its pre-induction level, with the immobile population decreased to 10.2% ($n = 1549$) and D of the mobile population increased to $2.87 \times 10^{-2} \pm 2.76 \times 10^{-3} \mu\text{m}^2/\text{s}$ ($n = 1391$) (Fig. 4B, 4C [↗](#)). Therefore, the PG-binding of LtgB shows a negative correlation with PG degradation. Taken together, LtgA and LtgB display opposite responses to glycerol induction, in which LtgB rapidly binds to PG before yielding to LtgA, whose PG-binding is concurrent with the rod-to-sphere transition. The sequential PG-binding by these two LTGs controls the pace of PG-degradation during glycerol-induced sporulation.

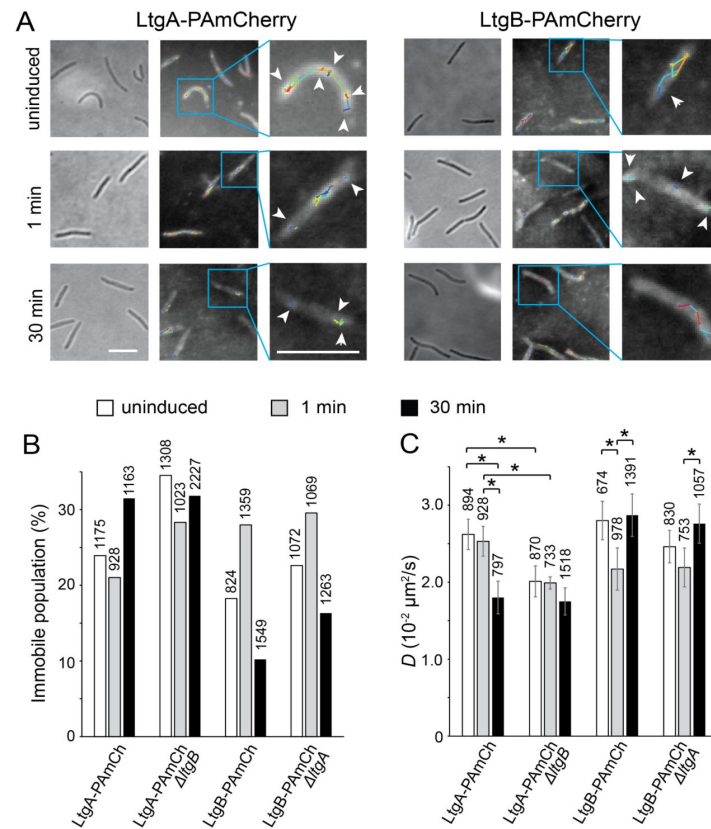


Fig. 4.

LtgB regulates the PG-binding of LtgA during glycerol-induced sporulation.

A) LtgB exhibits a response to glycerol induction earlier than LtgA. Representative trajectories of LtgA and LtgB before (uninduced) and after (1 min and 30 min) glycerol induction. The overall distribution of both LTGs is displayed using the composite of 100 consecutive frames taken at 100-ms intervals. Single-particle trajectories of PAmCherry were generated from the same frames. Individual trajectories are distinguished by colors. Scale bars, 5 μm . **B** and **C**) the absence of LtgB reduces the diffusion of LtgA, which is reflected in the increase of immobile population (**B**) and the decrease in D (**C**), and these effects are especially prominent in the cells before (uninduced) and immediately after (1 min) glycerol induction. For each protein and condition, particles were identified from at least 100 cells and three independent experiments. The total number of particles analyzed is shown on top of each plot. Error bars were the standard derivation of 1,000 bootstrap samples and * indicates a significant difference of > 0.005 .

LtgB blocks LtgA from binding PG in the early stage of glycerol-induced sporulation

Does LtgB's early PG-binding suppress PG degradation by LtgA? To answer this question, we expressed LtgA-PAmCherry using the native *ltgA* locus and promoter in the Δ *ltgB* background. In vegetative cells, the absence of LtgB significantly reduced the mobility of single LtgA-PAmCherry particles, suggesting that LtgB does affect LtgA's binding to the PG (Fig. 4B, 4C). Similarly, we expressed LtgB-PAmCherry using the native *ltgB* locus and promoter in the Δ *ltgA* background. In contrast, the mobility of single LtgB-PAmCherry particles only decreased slightly in the absence of LtgA (Fig. 4B, 4C). These results suggest that while LtgB significantly reduces LtgA's binding to PG, probably due to its higher expression level, LtgA has only a modest impact on LtgB's ability to bind PG.

Strikingly different from its slow response to glycerol in wild-type cells, LtgA increased its PG-binding immediately (1 min) after glycerol induction in the Δ *ltgB* background, confirming that LtgB restricts LtgA access to the PG in the early stage of sporulation (Fig. 4B, 4C). In contrast, at 30 min of induction, a time point when LtgB dissociated from PG, its absence no longer affected LtgA's PG-binding (Fig. 4B, 4C). Similar to the observation in wild-type cells, the lack of LtgA did not affect the PG-binding of LtgB at either 1 min or 30 min after glycerol induction (Fig. 4B, 4C). In summary, LtgB, a slow LTG, regulates the progression of rapid sporulation by preventing LtgA, a fast LTG, from excessively degrading PG, thereby ensuring that sporulating cells can deposit polysaccharide spore coats before PG disintegration occurs.

Upregulated PG synthesis negatively affects PG degradation

Since LtgA is less mobile in vegetative Δ *ltgB* cells, i.e. strongly binds to PG (Fig. 4B, C), why do these cells retain their rod shape rather than losing it spontaneously before glycerol induction? In addition to PG-binding, LtgA may require additional stimuli to initiate PG degradation. We hypothesized that, similar to certain antibiotics targeting PG synthases, which induce wild-type cells to degrade PG and form pseudospores in rich liquid media (Zhang *et al.*, 2023b, O'Connor & Zusman, 1997), glycerol may alter cellular metabolism, leading to reduced PG synthesis. If this is the case, cells with upregulated PG synthesis should remain rods even in the presence of glycerol.

To test our hypothesis, we used the vanillate-inducible promoter (Iniesta *et al.*, 2012) to overexpress *murA* as a merodiploid in the wild-type background. Because MurA catalyzes the first committed step of PG synthesis that produces UDP-MurNAc (Egan *et al.*, 2020, Rohs & Bernhardt, 2021), elevated MurA levels are expected to channel more cellular resources toward PG synthesis. Overproduction of MurA in the presence of 200 μ M vanillate resulted in a heterogeneous cell population, with normal cells coexisting alongside elongated ones (Fig. 2A, B). This heterogeneity may reflect the heterogeneous induction of *murA* across the whole population. Overall, excessive MurA increased the average length of vegetative cells by 19.6%. Cells overproducing MurA also displayed heterogeneity during glycerol-induced sporulation. While some cells transitioned into spheres, many retained their rod shape even after 6 h of induction (Fig. 2A, B). Notably, sporadic lysis of rod-shaped cells began 2 h post-induction and became increasingly frequent with continued incubation (Fig. 2A). Potentially, elevated muropeptide production may lead to the accumulation of toxic intermediates that the relatively insufficient LTGs failed to degrade (Weaver *et al.*, 2022). Nevertheless, the capacity of MurA overexpression to enable certain cells to circumvent glycerol-induced sporulation implies that PG production acts as a regulatory cue for PG degradation.

To test if excessive MurA can also reduce PG degradation during slow sporulation, we induced MurA overexpression on solid CF agar containing 100 μ M vanillate. After 96 h of incubation, these cells developed fruiting bodies that were noticeably larger and flatter compared to those of the wild-type strain (Fig. 2C). To test if such fruiting bodies contained mature spores, we scraped

them from the agar surface and tested the germination rate of spores on CYE agar after sonication. The fruiting bodies formed by cells overexpressing MurA yielded 21.3% of the colonies produced by an equivalent number of wild-type cells, indicating that upregulated PG synthesis also prevents slow sporulation. In contrast, cells grown without vanillate progressed normally through both sporulation pathways, showing no differences compared to wild-type cells (**Fig. S2**). Therefore, the sporulation of *M. xanthus* is sensitive to PG synthesis and increased PG synthesis balances PG degradation in both sporulation pathways.

Discussion

It is still unknown how many bacteria outside of the *Firmicutes* phylum form true spores. Nevertheless, our findings elucidate a new mechanism of non-firmicute bacterial sporulation that depends on PG degradation. This mechanism presents a significant challenge for sporulating cells: preserving their structural integrity while simultaneously dismantling the primary framework that upholds it. Our findings revealed that besides regulating the production of LTGs, cells could control the pace of PG degradation in a rapid sporulation pathway through the regulation between two LTGs and hence allow the resistant polysaccharide coats to assemble and mature.

Studying the regulation between two LTGs is especially challenging both *in vivo* and *in vitro* for a few reasons. First, it is difficult to monitor their instantaneous activities *in vivo*. Secondly, muropeptide analysis may not distinguish their roles because their products bear the same signature, anhydro-MurNAc. Thirdly, some enzymes require special substrates or activators, which would be difficult to study *in vitro*. To overcome these technical hurdles, we leveraged single-particle mobility to quantify the PG-binding of these enzymes. The simultaneous occurrence of reduced LtgA mobility and PG degradation during glycerol-induced sporulation indicates that the molecular dynamics of LtgA accurately mirrors its enzymatic activity. Such correlation between decreased particle mobility and increased enzymatic activity applies to many other PG-related enzymes, including multiple PG polymerases in *E. coli* and the endopeptidase DacB in *M. xanthus* (Lee *et al.*, 2016, Zhang *et al.*, 2023b, Yang *et al.*, 2021). Because the size difference between PG and its related enzymes is a conserved feature in all bacteria, our method using single-particle tracking to quantify PG-binding can be applied in many organisms.

We discovered that LtgB, a slower-acting LTG, limits the access of the faster LtgA to the PG during the early phase of rapid sporulation, thereby regulating the rate of its degradation. Then how does LtgA breach the blockage of LtgB and gain access to PG as sporulation progresses? Because the overexpression of LtgA is sufficient to damage PG in uninduced cells (**Fig. 3B**), we propose that the relative abundance of these two LTGs plays a critical role in determining the fate during fast sporulation. While LtgB nearly saturates PG binding sites in vegetative cells where few LtgA are produced, LtgA—upregulated after glycerol induction (Muller *et al.*, 2010)—gradually outcompetes LtgB and thereby causes LtgB to disassociate from PG.

As increased PG synthesis enables many cells to bypass both sporulation pathways, it is reasonable to propose that diminished PG synthesis initiates PG degradation by LTGs. While functional coordination between PG synthesis and hydrolysis has been speculated for over 50 years (Koch, 1985, Koch, 1990), its mechanisms only came to light recently. Some endopeptidases are proposed to serve as the space-makers for PG synthases (Dorr *et al.*, 2013, Singh *et al.*, 2012). In *M. xanthus*, we reported that moenomycin, an antibiotic that specifically binds to the GTase domains of class A penicillin-binding proteins (aPBPs) (Sung *et al.*, 2009, Lovering *et al.*, 2007), specifically activates a PG endopeptidase DacB through PBP1a2, an aPBP (Zhang *et al.*, 2023b). Because moenomycin mimics a growing glycan strand in the GTase domains of aPBPs, it actually locks aPBPs in their glycan-charged conformations (Sung *et al.*, 2009, Lovering *et al.*, 2007). Thus, similar to the moenomycin-bound form, glycan-charged PBP1a2 in physiological conditions recruits DacB to the PG assembly sites, which in turn, generates openings in the existing PG

network as the crosslink sites for the TPase activity of PBP1a2. Such “make-before-break” mechanism (Koch & Doyle, 1985 [DOI](#)) could prevent unneeded hydrolysis and thus maintain PG integrity.

Compared to the well-characterized space-making functions of endopeptidases, the roles of LTGs, beyond PG recycling, remain ill-defined. Even less is known on their potential coordination with PG synthases (Weaver *et al.*, 2023 [DOI](#)). Because LTGs from bacterial and phage origins are inhibited by L,D-crosslinks, they may functionally connect to PBPs that form D,D-crosslinks (Alvarez *et al.*, 2024 [DOI](#)). Additionally, some LTGs, such as MltD and MltG in *E. coli* and MltG in *Vibrio cholerae*, prefer to degrade nascent, uncrosslinked glycan strands (Kaul *et al.*, 2024 [DOI](#), Weaver *et al.*, 2022 [DOI](#), Bohrhunter *et al.*, 2021 [DOI](#)). Our recent report indicates that AgmT, an *M. xanthus* LTG homologous to *E. coli* MltG, detoxifies uncrosslinked glycan strands under antibiotic stress and modifies crosslinked PG scaffolds to attach a motility machinery to PG (Ramirez Carbo *et al.*, 2024 [DOI](#)). Then how does reduced PG synthesis activate LtgA in *M. xanthus*? Although MltE, the closest homolog of LtgA in *E. coli*, degrades glycan strands regardless of their crosslinking status or peptide content, it shows a marked preference for uncrosslinked substrates (Dik *et al.*, 2017 [DOI](#), Fibriansah *et al.*, 2012 [DOI](#)). Similar to AgmT and MltE, LtgA might also bind to both crosslinked and uncrosslinked glycan strands. Consistent with its high mobility, LtgA may coordinate with diffusive PG polymerases and cleave the uncrosslinked glycan strands. During rapid sporulation, as nascent glycan strands decline due to muropeptide depletion, LtgA increasingly binds to the crosslinked PG scaffold, which is evidenced by its reduced mobility. This binding intensifies further as PG degradation continues, establishing a positive feedback loop. In this case, LtgB serves as a brake that dampens this positive feedback, moderates PG degradation, and thus allows sporulating cells to maintain integrity by assembling polysaccharide coats. Such cross-regulation between PG hydrolases, which is currently under investigation, provides yet another mechanism by which bacteria dynamically maintain their PG cell walls.

Materials and methods

Bacterial strains and growth conditions

Vegetative *M. xanthus* cells were grown in liquid CYE medium (10 mM MOPS pH 7.6, 1% (w/v) Bacto™ casitone (BD Biosciences), 0.5% yeast extract and 8 mM MgSO₄) at 32 °C, in 125-ml flasks with vigorous shaking, or on CYE plates that contains 1.5% agar. We used strain DZ2 as the wild-type *M. xanthus* strain (Campos & Zusman, 1975 [DOI](#)). Knock-out mutants were constructed by electroporating DZ2 cells with 4 µg of plasmid DNA. Transformed cells were plated on CYE plates supplemented with 100 mg/ml sodium kanamycin sulfate or 10 mg/ml tetracycline. The strains and plasmids used in this study are listed in **Table 1** [DOI](#).

Sporulation and spore purification

Cells were grown in 25 ml liquid cell culture to OD₆₀₀ 0.8 – 1.2. Glycerol was added to 1 M to induce sporulation. The liquid culture was incubated at 32 °C with vigorous shaking. After 24 h, spores and cells were harvested by centrifugation (10 min, 10,000 g and 25 °C). To further purify the wild-type spores, the pellet was resuspended in 5 ml water, the remaining vegetative cells were eliminated by sonication (Cole Palmer 4710 ultrasonic homogenizer, 30% output, 10 cycles), and sonication-resistant spores were washed three times with water and collected by centrifugation (5 min, 10,000 g and 4 °C). The elimination of vegetative cells was confirmed by DIC microscopy.

For phenotypic assays on starvation-induced fruiting body formation, vegetative cells (10 µl), at a concentration of 4×10^9 colony-forming units (cfu) ml⁻¹, were spotted on CF (0.015% Casitone, 0.2% sodium citrate, 0.1% sodium pyruvate, 0.02% (NH₄)₂SO₄, 10 mM MOPS (pH 7.6), 8 mM MgSO₄ and 1

<i>M. xanthus</i> Strain	Description	Source
DZ2	The wild-type strain	(Campos & Zusman, 1975)
BN361	$\Delta ItgA$	This study
BN362	$\Delta ItgB$	This study
BN363	$\Delta ItgA \Delta ItgB$	This study
BN364	$\Delta ItgA$ pMR3629- <i>ItgB</i>	This study
BN365	DZ2 with P_{van} - <i>ItgA</i>	This study
BN366	DZ2 with P_{van} - <i>ItgB</i>	This study
BN367	DZ2 with P_{van} - <i>murA</i>	This study
BN368	<i>ItgA</i> -PAmCherry	This study
BN369	<i>ItgB</i> -PAmCherry	This study
BN370	$\Delta ItgB$ <i>ItgA</i> -PAmCherry	This study
BN371	$\Delta ItgA$ <i>ItgB</i> -PAmCherry	This study
<i>E. coli</i> Strain	Description	Source
DH5 α	<i>E. coli</i> strain for molecular cloning	lab collection
BL31 (DE3)	<i>E. coli</i> host for protein expression	Novagen
Plasmid name	Description	Source
pMR3629	Plasmid for overexpressing genes as merodiploids using a vanilate-induced promoter, tet ^R	(Iniesta <i>et al.</i> , 2012)
pMR3629- <i>ItgA</i>	P_{van} - <i>ItgA</i> in pMR3629, tet ^R	This study
pMR3629- <i>ItgB</i>	P_{van} - <i>ItgB</i> in pMR3629, tet ^R	This study
pMR3629- <i>murA</i>	P_{van} - <i>murA</i> in pMR3629, tet ^R	This study
pET28a- <i>ItgA</i>	Plasmid for overexpression of His-tagged <i>ItgA</i>	This study
pET28a- <i>ItgB</i>	Plasmid for overexpression of His-tagged <i>ItgB</i>	This study

Table 1.

Strains and plasmids used in this study.

mM KH_2PO_4) plates containing an agar concentration of 1.5%, incubated at 32°C. To purify fruiting body spores, cells were grown in 25 ml liquid cell culture to OD_{600} 0.8 – 1.2 and harvested by centrifugation (10 min, 10,000 g and 25 °C). The pellet was resuspended in 1 ml water and plated on two 150 mm CF plates containing 1.5% agar using a spreader. The plates were air-dried and incubated at 32 °C for 120 h. For the wild-type and ΔltgA strains, fruiting bodies were scraped from the plates, suspended in 2 ml water, and subjected to sonication (Cole Palmer 4710 ultrasonic homogenizer, 60% output, 20 cycles). Spores were wash three times in water and collected by centrifugation (5 min, 10,000 g and 4 °C). For the ΔltgB strains, the sonication step was omitted.

Peptidoglycan purification and UPLC analysis

For vegetative cell peptidoglycan analysis, samples were processed as previously described for Gram negative bacteria (Alvarez *et al.*, 2016 [↗](#), Desmarais *et al.*, 2013 [↗](#)). In brief, *M. xanthus* cells were grown until mid-stationary phase and harvested by centrifugation (30 min, 8,000 g). Supernatant was discarded and the pellet was resuspended and boiled in 1x PBS with 5% SDS for 2 h. The saccules were repeatedly washed with MilliQ water by ultracentrifugation (150,000 g, 10 min, 20°C) to remove the remaining SDS. The samples were then treated with muramidase (100 µg/mL) for 16 hours at 37 °C. Muramidase digestion was stopped by boiling and coagulated proteins were removed by centrifugation (10 min, 15,000 g). The supernatants were first adjusted to pH 8.5-9.0 with sodium borate buffer and then sodium borohydride was added to a final concentration of 10 mg/mL. After reducing the samples at room temperature for 30 mins, the pH was adjusted to pH 3.5 with orthophosphoric acid.

UPLC analyses of muropeptides were performed on a Waters UPLC system (Waters Corporation, USA) equipped with an ACQUITY UPLC BEH C18 Column, 130Å, 1.7 µm, 2.1 mm X 150 mm (Waters, USA) and a dual wavelength absorbance detector. Elution of muropeptides was detected at 204 nm. Muropeptides were separated at 45 °C using a linear gradient from buffer A (formic acid 0.1% in water) to buffer B (formic acid 0.1% in acetonitrile) in a 25-min run, with a 0.30 mL/min flow. For vegetative cells analysis, relative total PG amounts were calculated by comparison of the total intensities of the chromatograms (total area) from three biological replicas normalized to the same initial biomass and extracted with the same volumes. Quantification of muropeptides was based on their relative abundances (relative area of the corresponding peak).

Imaging and data analysis

For all imaging experiments on isolated spores/cells, we spotted 5 µl of cells grown in liquid CYE medium to $\text{OD}_{600} \sim 1$ on agar (1.5%) pads and imaged using a Andor iXon Ultra 897 EMCCD camera (effective pixel size 160 nm) on an inverted Nikon Eclipse-Ti™ microscope with a 100× 1.49 NA TIRF objective. Fruiting bodies were photographed after a 120-h incubation on CF agar using a Nikon SMZ1000 microscope and an OMAX A3590U digital camera. Cell morphology was imaged at indicated time points using DIC microscopy. The geometric aspect ratios (L/W) of spores/cells were determined using a custom algorithm written in MATLAB, which is available in the GitHub repository, https://github.com/NanLabMyxo/Rod_shape_paper [↗](#) (Zhang *et al.*, 2020 [↗](#), Zhang *et al.*, 2023a [↗](#), Zhang *et al.*, 2023b [↗](#)).

For sptPALM, *M. xanthus* cells were grown in CYE to 4×10^8 cfu/ml, spotted on 1.5% agar pads and subjected to highly inclined and laminated optical sheet (HILO) illumination (Fu *et al.*, 2018 [↗](#), Nan *et al.*, 2015 [↗](#), Nan *et al.*, 2013 [↗](#), Tokunaga *et al.*, 2008 [↗](#)). PAmCherry was activated using a 405-nm laser (0.3 kW/cm²), excited and imaged using a 561-nm laser (0.2 kW/cm²). Images were acquired at 10 Hz. For each sptPALM experiment, single PAmCherry particles were localized in at least 100 individual cells from three biological replicates. sptPALM data were analyzed using a MATLAB (MathWorks) script, which is available in the GitHub repository, https://github.com/NanLabMyxo/Rod_shape_paper [↗](#) (Zhang *et al.*, 2020 [↗](#), Zhang *et al.*, 2023a [↗](#), Zhang *et al.*, 2023b [↗](#)). Briefly, cells were identified using differential interference contrast images. Single PAmCherry particles inside cells were fit by a symmetric 2D Gaussian function, whose center was

assumed to be the particle's position (Fu *et al.*, 2018). Particles that explored areas smaller than 160 nm × 160 nm (within one pixel) in 0.4 – 1.2 s were considered immotile (Fu *et al.*, 2018, Zhang *et al.*, 2023a, Zhang *et al.*, 2023b). *D* of all the mobile particles was determined from a linear fit to the first four points of the MSD using a formula $MSD = 4D\Delta t$ (Fu *et al.*, 2018, Lee *et al.*, 2016). Error bars were the standard derivation of 1,000 bootstrap samples using the published method (Morgenstein *et al.*, 2015). Sample trajectories were generated using the TrackMate (Ershov *et al.*, 2022) plugin in the ImageJ suite (<https://imagej.net>).

Immunofluorescence

Cell samples were resuspended and boiled in 1x PBS with 5% sodium dodecyl sulfate (SDS) for 2 h. The saccules were repeatedly washed with water by centrifugation (5 min, 15,000 g and 25 °C) to remove the remaining SDS. Remaining proteins were further removed by incubation with proteinase K (1 µg/ml) for 2 h at 37 °C. Microscope cover slides were prepared by coating with 100 µl 0.1% poly-L-lysine for 5 min at room temperature (RT), washing with ddH₂O and air drying. 100 µl samples were spotted onto the slides and incubated for 20 min at RT. The slides were then washed three times with phosphate buffer saline (PBS; 81 mM Na₂HPO₄, 15 mM KH₂PO₄, 1.37 M NaCl, 27 mM KCl, pH 7.4), and then incubated with anti-PG primary antibody¹⁶ diluted 1:1000 in PBS at RT for 1 h. Slides were washed 6 times with PBS and then incubated with Alexa Fluor 488 conjugated goat-anti-rabbit secondary antibody (Invitrogen) at a 1:1000 dilution in PBS at RT for 1 h in the dark. Slides were washed 6 times with PBS. Each cover glass was applied to a supporting slide, fixed with nail polish, and stored at 4 °C, if necessary. Images were recorded using a 488-nm laser.

Protein expression and purification

DNA sequences encoding amino acids 21 – 243 of LtgA and 26 – 710 of LtgB were amplified by polymerase chain reaction (PCR) and inserted into the pET28a vector (Novagen) between the restriction sites of *Eco*RI and *Hind*III and used to transform *E. coli* strain BL21(DE3). Transformed cells were cultured in 20 ml LB (Luria-Bertani) broth at 37 °C overnight and used to inoculate 1 L LB medium supplemented with 1.0% glucose. Protein expression was induced by 0.1 mM IPTG (isopropyl- β -D-thiogalactopyranoside) when the culture reached an OD₆₀₀ of 0.8. Cultivation was continued at 16 °C for 10 h before the cells were harvested by centrifugation at 6,000 × g for 20 min. Proteins were purified using a NGC™ Chromatography System (BIO-RAD) and 5-ml HisTrap™ columns (Cytiva) (Pogue *et al.*, 2018, Nan *et al.*, 2010). Purified proteins were concentrated using Amicon™ Ultra centrifugal filter units (Millipore Sigma) with a 10-kDa molecular weight cutoff and stored at –80 °C.

LTG activity (RBB) assay

PG was purified following the published method (Zhang *et al.*, 2023b, Alvarez *et al.*, 2016, Ramirez Carbo *et al.*, 2024). In brief, *M. xanthus* cells were grown until mid-stationary phase and harvested by centrifugation (6,000 × g, 20 min, 25 °C). Supernatant was discarded and the pellet was resuspended and boiled in 1× PBS with 5% SDS for 2 h. SDS was removed by repetitive wash with water and centrifugation (21,000 × g, 10 min, 25 °C). Purified PG from 100 ml culture was suspended into 1 ml 1× PBS and stored at –20 °C. RBB labelling of PG was performed essentially as previously described (Uehara *et al.*, 2010, Jorgenson *et al.*, 2014). Purified sacculi were incubated with 20 mM RBB in 0.25 M NaOH overnight at 37 °C. Reactions were neutralized by adding equal volumes of 0.25 M HCl and RBB-labeled PG was collected by centrifugation at 21,000 × g for 15 min. Pellets were washed repeatedly with water until the supernatants became colorless. RBB-labelled sacculi were incubated with purified LtgA and LtgB (1 mg/ml) at 25 °C for 12 h. lysozyme (1 mg/ml) was used as a positive control. Dye release was quantified by the absorption at 595 nm from the supernatants after centrifugation (21,000 × g, 10 min, 25 °C).

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Additional files

Fig. S1, S2 

References

1. Alvarez L., Hernandez S.B., de Pedro M.A., Cava F. (2016) **Ultra-Sensitive, High-Resolution Liquid Chromatography Methods for the High-Throughput Quantitative Analysis of Bacterial Cell Wall Chemistry and Structure** *Methods Mol Biol* **1440**:11–27 [Google Scholar](#)
2. Alvarez L., Hernandez S.B., Torrens G., Weaver A.I., Dorr T., Cava F (2024) **Control of bacterial cell wall autolysins by peptidoglycan crosslinking mode** *Nature communications* **15**:7937 [Google Scholar](#)
3. Aramayo R., Nan B (2022) **De Novo Assembly and Annotation of the Complete Genome Sequence of *Myxococcus xanthus* DZ2** *Microbiol Resour Announc* :e0107421 [Google Scholar](#)
4. Bohrhunter J.L., Rohs P.D.A., Torres G., Yunck R., Bernhardt T.G (2021) **MltG activity antagonizes cell wall synthesis by both types of peptidoglycan polymerases in *Escherichia coli*** *Mol Microbiol* **115**:1170–1180 [Google Scholar](#)
5. Bui N.K., Gray J., Schwarz H., Schumann P., Blanot D., Vollmer W (2009) **The peptidoglycan sacculus of *Myxococcus xanthus* has unusual structural features and is degraded during glycerol-induced myxospore development** *J Bacteriol* **191**:494–505 [Google Scholar](#)
6. Campos J.M., Zusman D.R (1975) **Regulation of Development in *Myxococcus-xanthus* – Effect of 3' Doublebond 5'-Cyclic Amp, Adp, and Nutrition** *Proceedings of the National Academy of Sciences of the United States of America* **72**:518–522 [Google Scholar](#)
7. de Pedro M.A., Quintela J.C., Holtje J.V., Schwarz H. (1997) **Murein segregation in *Escherichia coli*** *J Bacteriol* **179**:2823–2834 [Google Scholar](#)
8. Desmarais S.M., De Pedro M.A., Cava F., Huang K.C. (2013) **Peptidoglycan at its peaks: how chromatographic analyses can reveal bacterial cell wall structure and assembly** *Mol Microbiol* **89**:1–13 [Google Scholar](#)
9. Dik D.A., Marous D.R., Fisher J.F., Mobashery S (2017) **Lytic transglycosylases: concinnity in concision of the bacterial cell wall** *Crit Rev Biochem Mol Biol* **52**:503–542 [Google Scholar](#)
10. Dorr T., Cava F., Lam H., Davis B.M., Waldor M.K (2013) **Substrate specificity of an elongation-specific peptidoglycan endopeptidase and its implications for cell wall architecture and growth of *Vibrio cholerae*** *Mol Microbiol* **89**:949–962 [Google Scholar](#)
11. Dworkin M., Gibson S.M (1964) **A System for Studying Microbial Morphogenesis: Rapid Formation of Microcysts in *Myxococcus xanthus*** *Science* **146**:243–244 [Google Scholar](#)
12. Egan A.J.F., Errington J., Vollmer W (2020) **Regulation of peptidoglycan synthesis and remodelling** *Nat Rev Microbiol* **18**:446–460 [Google Scholar](#)
13. Ershov D., Phan M.S., Pylvanainen J.W., Rigaud S.U., Le Blanc L., Charles-Orszag A., Conway J.R.W., Laine R.F., Roy N.H., Bonazzi D., Dumenil G., Jacquemet G., Tinevez J.Y. (2022) **TrackMate**

7: integrating state-of-the-art segmentation algorithms into tracking pipelines *Nat Methods* **19**:829–832 [Google Scholar](#)

14. Fibriansah G., Gliubich F.I., Thunnissen A.M (2012) **On the mechanism of peptidoglycan binding and cleavage by the endo-specific lytic transglycosylase MltE from Escherichia coli** *Biochemistry* **51**:9164–9177 [Google Scholar](#)
15. Fu G., Bandaria J.N., Le Gall A.V., Fan X., Yildiz A., Mignot T., Zusman D.R., Nan B. (2018) **MotAB-like machinery drives the movement of MreB filaments during bacterial gliding motility** *Proc Natl Acad Sci U S A* **115**:2484–2489 [Google Scholar](#)
16. Higgins D., Dworkin J (2012) **Recent progress in Bacillus subtilis sporulation** *FEMS Microbiol Rev* **36**:131–148 [Google Scholar](#)
17. Holkenbrink C., Hoiczky E., Kahnt J., Higgs P.I (2014) **Synthesis and assembly of a novel glycan layer in Myxococcus xanthus spores** *J Biol Chem* **289**:32364–32378 [Google Scholar](#)
18. Huang M., Hull C.M (2017) **Sporulation: how to survive on planet Earth (and beyond)** *Curr Genet* [Google Scholar](#)
19. Hutchison E.A., Miller D.A., Angert E.R. (2014) **Sporulation in Bacteria: Beyond the Standard Model** *Microbiol Spectr* **2** [Google Scholar](#)
20. Iniesta A.A., Garcia-Heras F., Abellon-Ruiz J., Gallego-Garcia A., Elias-Arnanz M (2012) **Two systems for conditional gene expression in Myxococcus xanthus inducible by isopropyl-beta-D-thiogalactopyranoside or vanillate** *J Bacteriol* **194**:5875–5885 [Google Scholar](#)
21. Jorgenson M.A., Chen Y., Yahashiri A., Popham D.L., Weiss D.S (2014) **The bacterial septal ring protein RlpA is a lytic transglycosylase that contributes to rod shape and daughter cell separation in Pseudomonas aeruginosa** *Mol Microbiol* **93**:113–128 [Google Scholar](#)
22. Kaul M., Meher S.K., Nallamotu K.C., Reddy M (2024) **Glycan strand cleavage by a lytic transglycosylase, MltD contributes to the expansion of peptidoglycan in Escherichia coli** *PLoS Genet* **20**:e1011161 [Google Scholar](#)
23. Koch A.L (1985) **How Bacteria Grow and Divide in Spite of Internal Hydrostatic-Pressure** *Canadian Journal of Microbiology* **31**:1071–1084 [Google Scholar](#)
24. Koch A.L (1990) **Additional arguments for the key role of “smart” autolysins in the enlargement of the wall of gram-negative bacteria** *Res Microbiol* **141**:529–541 [Google Scholar](#)
25. Koch A.L., Doyle R.J (1985) **Inside-to-outside growth and turnover of the wall of gram-positive rods** *J Theor Biol* **117**:137–157 [Google Scholar](#)
26. Kottel R.H., Bacon K., Clutter D., White D (1975) **Coats from Myxococcus xanthus: characterization and synthesis during myxospore differentiation** *J Bacteriol* **124**:550–557 [Google Scholar](#)
27. Lee T.K., Meng K., Shi H., Huang K.C (2016) **Single-molecule imaging reveals modulation of cell wall synthesis dynamics in live bacterial cells** *Nature communications* **7**:13170 [Google Scholar](#)

28. Lovering A.L., de Castro L.H., Lim D., Strynadka N.C. (2007) **Structural insight into the transglycosylation step of bacterial cell-wall biosynthesis** *Science* **315**:1402–1405 [Google Scholar](#)
29. McKenney P.T., Eichenberger P (2012) **Dynamics of spore coat morphogenesis in *Bacillus subtilis*** *Mol Microbiol* **83**:245–260 [Google Scholar](#)
30. Morgenstein R.M., Bratton B.P., Nguyen J.P., Ouzounov N., Shaevitz J.W., Gitai Z (2015) **RodZ links MreB to cell wall synthesis to mediate MreB rotation and robust morphogenesis** *Proc Natl Acad Sci U S A* [Google Scholar](#)
31. Muller F.D., Schink C.W., Hoiczky E., Cserti E., Higgs P.I (2012) **Spore formation in *Myxococcus xanthus* is tied to cytoskeleton functions and polysaccharide spore coat deposition** *Mol Microbiol* **83**:486–505 [Google Scholar](#)
32. Muller F.D., Treuner-Lange A., Heider J., Huntley S.M., Higgs P.I (2010) **Global transcriptome analysis of spore formation in *Myxococcus xanthus* reveals a locus necessary for cell differentiation** *BMC Genomics* **11**:264 [Google Scholar](#)
33. Munoz-Dorado J., Moraleda-Munoz A., Marcos-Torres F.J., Contreras-Moreno F.J., Martin-Cuadrado A.B., Schrader J.M., Higgs P.I., Perez J (2019) **Transcriptome dynamics of the *Myxococcus xanthus* multicellular developmental program** *eLife* **8** [Google Scholar](#)
34. Nan B., Bandaria J.N., Guo K.Y., Fan X., Moghtaderi A., Yildiz A., Zusman D.R (2015) **The polarity of myxobacterial gliding is regulated by direct interactions between the gliding motors and the Ras homolog MglA** *Proc Natl Acad Sci U S A* **112**:E186–193 [Google Scholar](#)
35. Nan B., Bandaria J.N., Moghtaderi A., Sun I.H., Yildiz A., Zusman D.R (2013) **Flagella stator homologs function as motors for myxobacterial gliding motility by moving in helical trajectories** *Proc Natl Acad Sci U S A* **110**:E1508–1513 [Google Scholar](#)
36. Nan B., Liu X., Zhou Y., Liu J., Zhang L., Wen J., Zhang X., Su X.D., Wang Y.P (2010) **From signal perception to signal transduction: ligand-induced dimeric switch of DctB sensory domain in solution** *Mol Microbiol* **75**:1484–1494 [Google Scholar](#)
37. O'Connor K.A., Zusman D.R (1988) **Reexamination of the role of autolysis in the development of *Myxococcus xanthus*** *J Bacteriol* **170**:4103–4112 [Google Scholar](#)
38. O'Connor K.A., Zusman D.R (1997) **Starvation-independent sporulation in *Myxococcus xanthus* involves the pathway for beta-lactamase induction and provides a mechanism for competitive cell survival** *Mol Microbiol* **24**:839–850 [Google Scholar](#)
39. Perez-Burgos M., Garcia-Romero I., Valvano M.A., Sogaard Andersen L (2020) **Identification of the Wzx flippase, Wzy polymerase and sugar-modifying enzymes for spore coat polysaccharide biosynthesis in *Myxococcus xanthus*** *Mol Microbiol* **113**:1189–1208 [Google Scholar](#)
40. Pogue C.B., Zhou T., Nan B (2018) **PlpA, a PilZ-like protein, regulates directed motility of the bacterium *Myxococcus xanthus*** *Mol Microbiol* **107**:214–228 [Google Scholar](#)
41. Popham D.L., Bernhards C.B (2015) **Spore Peptidoglycan** *Microbiol Spectr* **3** [Google Scholar](#)
42. Ramirez Carbo C.A., Faromiki O.G., and Nan B. (2024) **A lytic transglycosylase connects bacterial focal adhesion complexes to the peptidoglycan cell wall** *eLife* **13** [Google Scholar](#)

43. Rohs P.D.A., Bernhardt T.G (2021) **Growth and Division of the Peptidoglycan Matrix** *Annu Rev Microbiol* **75**:315–336 [Google Scholar](#)
44. Saidi F., Jolivet N.Y., Lemon D.J., Nakamura A., Belgrave A.M., Garza A.G., Veyrier F.J., Islam S.T (2021) **Bacterial glycocalyx integrity drives multicellular swarm biofilm dynamism** *Mol Microbiol* **116**:1151–1172 [Google Scholar](#)
45. Singh S.K., SaiSree L., Amrutha R.N., Reddy M (2012) **Three redundant murein endopeptidases catalyse an essential cleavage step in peptidoglycan synthesis of Escherichia coli K12** *Mol Microbiol* **86**:1036–1051 [Google Scholar](#)
46. Sung M.T., Lai Y.T., Huang C.Y., Chou L.Y., Shih H.W., Cheng W.C., Wong C.H., Ma C (2009) **Crystal structure of the membrane-bound bifunctional transglycosylase PBP1b from Escherichia coli** *Proc Natl Acad Sci U S A* **106**:8824–8829 [Google Scholar](#)
47. Tocheva E.I., Matson E.G., Morris D.M., Moussavi F., Leadbetter J.R., Jensen G.J (2011) **Peptidoglycan remodeling and conversion of an inner membrane into an outer membrane during sporulation** *Cell* **146**:799–812 [Google Scholar](#)
48. Tokunaga M., Imamoto N., Sakata-Sogawa K (2008) **Highly inclined thin illumination enables clear single-molecule imaging in cells** *Nat Methods* **5**:159–161 [Google Scholar](#)
49. Uehara T., Parzych K.R., Dinh T., Bernhardt T.G (2010) **Daughter cell separation is controlled by cytokinetic ring-activated cell wall hydrolysis** *EMBO J* **29**:1412–1422 [Google Scholar](#)
50. van Heijenoort J. (2011) **Peptidoglycan hydrolases of Escherichia coli** *Microbiol Mol Biol Rev* **75**:636–663 [Google Scholar](#)
51. Voelz H., Dworkin M (1962) **Fine structure of Myxococcus xanthus during morphogenesis** *J Bacteriol* **84**:943–952 [Google Scholar](#)
52. Waite D.W., Chuvochina M., Pelikan C., Parks D.H., Yilmaz P., Wagner M., Loy A., Naganuma T., Nakai R., Whitman W.B., Hahn M.W., Kuever J., Hugenholtz P (2020) **Proposal to reclassify the proteobacterial classes Deltaproteobacteria and Oligoflexia, and the phylum Thermodesulfobacteria into four phyla reflecting major functional capabilities** *Int J Syst Evol Microbiol* **70**:5972–6016 [Google Scholar](#)
53. Wartel M., Ducret A., Thutupalli S., Czerwinski F., Le Gall A.V., Mauriello E.M., Bergam P., Brun Y.V., Shaevitz J., Mignot T. (2013) **A versatile class of cell surface directional motors gives rise to gliding motility and sporulation in Myxococcus xanthus** *PLoS Biol* **11**:e1001728 [Google Scholar](#)
54. Weaver A., Taguchi A., Dorr T (2023) **Masters of Misdirection: Peptidoglycan Glycosidases in Bacterial Growth** *J Bacteriol* **205**:e0042822 [Google Scholar](#)
55. Weaver A.I., Alvarez L., Rosch K.M., Ahmed A., Wang G.S., van Nieuwenhze M.S., Cava F., Dorr T. (2022) **Lytic transglycosylases mitigate periplasmic crowding by degrading soluble cell wall turnover products** *eLife* **11** [Google Scholar](#)
56. Williams A.H., Wheeler R., Rateau L., Malosse C., Chamot-Rooke J., Haouz A., Taha M.K., Boneca I.G (2018) **A step-by-step in crystallo guide to bond cleavage and 1,6-anhydro-sugar**

product synthesis by a peptidoglycan-degrading lytic transglycosylase *J Biol Chem* **293**:6000–6010 [Google Scholar](#)

57. Yang X., McQuillen R., Lyu Z., Phillips-Mason P., De La Cruz A., McCausland J.W., Liang H., DeMeester K.E., Santiago C.C., Grimes C.L., de Boer P., Xiao J. (2021) **A two-track model for the spatiotemporal coordination of bacterial septal cell wall synthesis revealed by single-molecule imaging of FtsW** *Nat Microbiol* **6**:584–593 [Google Scholar](#)
58. Yutin N., Galperin M.Y (2013) **A genomic update on clostridial phylogeny: Gram-negative spore formers and other misplaced clostridia** *Environ Microbiol* **15**:2631–2641 [Google Scholar](#)
59. Zhang H., Mulholland G.A., Seef S., Zhu S., Liu J., Mignot T., Nan B (2020) **Establishing rod shape from spherical, peptidoglycan-deficient bacterial spores** *Proc Natl Acad Sci U S A* **117**:14444–14452 [Google Scholar](#)
60. Zhang H., Venkatesan S., Nan B.Y (2021) **Myxococcus xanthus as a Model Organism for Peptidoglycan Assembly and Bacterial Morphogenesis** *Microorganisms* **9** [Google Scholar](#)
61. Zhang H., Venkatesan S., Ng E., Nan B. (2023a) **Coordinated peptidoglycan synthases and hydrolases stabilize the bacterial cell wall** *GitHub*
62. Zhang H., Venkatesan S., Ng E., Nan B (2023b) **Coordinated peptidoglycan synthases and hydrolases stabilize the bacterial cell wall** *Nature communications* **14**:5357 [Google Scholar](#)

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Reviewer #1 (Public review):

Summary:

Ramirez Carbo et al. use the powerful *M. xanthus* spore morphogenesis model to address fundamental mechanisms in coordinated peptidoglycan remodeling and degradation. As peptidoglycan is an essential macromolecule and difficult to study in vivo, the authors use indirect but important methodology. The authors first identify two lytic transglycosylase (Ltg) enzymes necessary for spore morphogenesis using mutant phenotypic studies. They characterize these mutants for their role in coordinating spore morphogenesis induced either in fruiting bodies (starvation-dependent) or in liquid-rich media conditions (chemical-dependent). They conclude from these phenotypic and epistatic analyses that LtgA is necessary for morphogenesis during chemical-induced sporulation, and LtgB appears to be necessary to coordinate LtgA activity by interfering with LtgA function. Under starvation-induced sporulation, the absence of LtgB interferes with the building of fruiting bodies. LtgA does not appear to play a primary role in promoting aggregation into fruiting bodies, nor in degradation of peptidoglycan as assayed by loss of signal in anti-PG immunofluorescence. The authors demonstrate that the purified periplasmic domain of LtgA is highly active in degrading purified PG sacculi in vitro, while that of LtgB is highly reduced (relative to LtgA or lysozyme). The authors use photoactivated mCherry Lyt fusions and PALM to track the fusion protein mobility, which they state correlates with activity as immobilization results from PG binding. They demonstrate that in vegetative cells, a greater proportion of LtgA-PAmCh is more immobile (more active) than LtgB-PAmCh, but that directly after chemical-induction of sporulation, LtgB-PAmCh becomes more immobile (active). These analyses in the partner mutant backgrounds suggest that LtgA-PAmCh is more immobile (less active) in the absence of LtgB, but the reverse is not observed. Finally, the authors demonstrate that overexpression

of LtgA in vegetative conditions leads to cell rounding, likely because of uncontrolled PG degradation, while overexpression of LtgB displays no phenotype.

Strengths:

This paper capitalizes on a novel spore morphogenesis mechanism to define proteins and mechanisms involved in peptidoglycan reorganization. The authors use the powerful PALM microscopy technique to assess Ltg activity in vivo by assaying for immobility as a proxy for PG binding. The authors elucidate a novel mechanism by which two Ltg's function together - with one (LtgB) seeming to regulate the activity of the other (the primary Ltg).

Despite some weaknesses, there is no question that this study provides important insight into mechanisms of peptidoglycan remodeling - a difficult but highly impactful area of study with implications for the development of novel therapeutics and the discovery of mechanisms of fundamental bacterial physiology.

Weaknesses:

In many places, the authors do not adequately justify interpretations of their assays, leading to some apparently unjustified conclusions. Many of these are minor and may just require citations to demonstrate that the interpretations are justified by previous studies (detailed in recommendations below), but two bigger concerns are as follows:

(1) It is not clear how the muropeptides listed in Figure 1 were assigned, and it is missing in the methods. In the sporulating conditions, the spectra look like combinations of multiple peaks, and the data, as stated, is not convincing to the non-specialist eye.

(2) The observation that the *lytB* mutant prevents appropriate aggregation into fruiting bodies does not allow the interpretation that the absence of *LytB* prevents PG morphogenesis in the starvation-induced sporulation pathway, per se. It is more likely that in the *lytB* mutant, the morphogenesis program is not even triggered. This is because signaling proteins and regulators (specifically, C-signal accumulation/activated FruA), which are dependent on increased cell-cell signaling in the fruiting body, do not accumulate appropriately in shallow aggregates. C-signal/FruA are necessary to trigger the sporulation program in FBs. BTW: A hypothesis to explain the indirect effect of *ltgB* absence on aggregation could be that UDP-precursors are not regulated appropriately (unregulated *LtyA*??), so polysaccharides necessary for motility are not properly produced.

Along these lines, fruiting body formation does not equal sporulation, and even "darkened" fruiting bodies can be misleading, as some mutants form polysaccharide-rich fruiting bodies (that appear dark under certain light conditions in the stereomicroscope) but do not sporulate efficiently. The wording in the text suggests that the authors assume that sporulation levels are normal because fruiting bodies are produced (see specific comments for details).

(3) The authors repeatedly state that production of spore coat polysaccharides likely affects the PG IP staining (see below), but this is not well justified. A citation is needed if this has already been directly shown, or the language needs to be softened.

(4) Better justification for the immobility of *Lyt* proteins in vivo as an assay for activity may be required. If this is well known in the field, it should be explicitly stated. The authors address this better in the discussion - but still state it is a correlation.

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Reviewer #2 (Public review):**Summary:**

The authors' initial goal was to demonstrate loss of PG during the slow sporulation process of *Myxococcus xanthus*, with examination of the PG degradation products in order to implicate possible enzymes involved. Upon finding a predominance of LGT products, they examined sporulation in strains lacking each of the 14 candidate LGTs encoded in the genome, leading to the identification of two sporulation-linked LGTs. An extensive characterization of the roles played by these LGTs. One LGT is responsible for the slow sporulation PG degradation, while another is required for the rapid sporulation process. Interestingly, the "slow" LGT seems to provide an important regulatory brake on the rapid enzyme. Single-molecule fluorescent tracking of these enzymes was used to develop a model for their interaction with PG that mimics their observed activity. The rate of PG synthesis activity was also shown to impact the rate of PG degradation, suggesting potential interplay between the synthetic and degradative enzymes.

Strengths:

The genetic analysis to identify sporulation-linked LGTs and their effects on growth, sporulation, and spore properties was well done and productive. The fluorescence microscopy to track LGT mobility, presumably tied to activity, produced a convincing argument about the mechanism of regulation of one LGT by another.

Weaknesses:

While the impact of LGTs on sporulation was clearly demonstrated, the PG analysis that resulted from the study of LGTs raised some important unanswered questions. The analyses suggest that the PG is degraded to quite small fragments, which would normally be lost during the purification of PG. How these small fragments were thus detected is unclear, and this suggests a more complex story concerning PG metabolism during sporulation. An anti-PG antibody is used to quantify PG in the spores, but it is not made clear what the specificity of this antibody is, and thus whether it would recognize the LGT-altered PG of the spore. The authors suggest a "new mechanism of sporulation" when they have actually simply identified an important factor (PG degradation by LGTs) within a complex "process of sporulation".

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